CSD 598 - Winter 2026

From visualisation to inferential statistics
Was: Metalevel look at CSD 501

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Last time: Data exploration

Any questions? Any thoughts?

This time: From visualisation to inferential statistics

- Choosing a graph
- Data exploration example
- Causality – to make sense of statistical dependence, correlation, and causality
- Inferential statistics
- Data analysis example

Resources for choosing a graph

- `https://royal-statistical-society.github.io/datavisguide/`
  Good, approachable guide to data visualisation.
- `https://www.data-to-viz.com/`
  Actual recommendations with actual code (R, Python, D3.js, React) to get going.
- `https://r-graph-gallery.com/`
  Neat gallery for getting to know different graph types. (Not exhaustive though.)
- Doesn't take much to find more. Please post any good ones you like in Canvas.
- My own favourite tool is the pairs plot in R. So we'll have a look at an example of that: `./RTpairs.pdf`.

Exploration example

- This comes from the latest data analysis I have run.
- As it is unpublished, I ask it not be recorded, and hence will not describe it here on the slides.
- I am happy to talk about it in detail and show how I've done the whole thing.

Preliminary to inference: Causality

Causality

- We all know – or should know – by now that correlation does not imply causation.
- But what is causality?
  - David Hume has said that causality can not be proven empirically because we can not know if the world changes at a fundamental level.
  - Immanuel Kant has replied to this that causality is not a feature of the universe, but rather a way or tool of reasoning.
  - Empirically, some phenomena in quantum physics break causality. And in the general theory of relativity we can not say what the order of two events occurring is because it depends on the point of observation.
  - And this could all be wrong, but it's unlikely (or seems unlikely given the evidence).
- Unless this was not news, it should make one think. And that's what I'd like to encourage in any case.

Causality and other tools

- It is not enough to know how to use a tool, we should also understand enough of its workings so that we know
  - the limits of our understanding and
  - the limits of the tool.
- Statistics is a tool. Like causality.

Inferential statistics

Types of inferential statistics

Broadly speaking we have

- Tests which generally say 'statistically this claim seems believable' or '...does not seem believable'.
- Models are generally predictive: Change this variable like so and that variable tends to change like so.
- And as usual the division between tests and models is fuzzy-ish.
- In other words, a test and a model can be the same thing.
- And in addition we have different types of statistics, but that is for another day.

More on tests

- Non-parametric tests: "This group tends to be larger than that group".
- Parametric tests: "This group has a larger expected value than that group."
- There are **a lot** of both types of tests.

More on models

- Models predict values of a variable (or a group of variables).
  - The predicting variable are usually called independent variables.
  - The predicted variable(s) is(are) usually called dependent variable(s).
- Significance of a model can often be tested or approximately evaluated, but this becomes more difficult with model complexity.
- In general, added complexity can easily lead to a model that is very difficult to understand.
- Visualisations help in understanding. They also help in diagnosis.

Model is a test, test is a model – sometimes

- ANOVA can be seen as a single predictor (controlled variable) linear regression model where the only predictor is a categorical variable.
- Simple regression – line fitting – can be seen as another way of testing for linear (Pearson's) correlation.

Model diagnosis – this will likely be moved to next time

- Collinearity (linear correlation) of the independent/predicting variables is bad (mathematically).
  - It can usually be spotted fairly easily by calculating correlations among the predictors and/or making scatter plots like a pairs plot.
- Heteroscedasticity will mess more the efficiency of prediction than the accuracy of it.
- Outliers can mess prediction accuracy.
- Here a bit of data exploration and model diagnostic plots go a long way:
  - Neat explanation of diagnostic plots can be found here `https://library.virginia.edu/data/articles/diagnostic-plots`
  - Collinearity can be seen in for example a pairs plot or a correlation matrix (not to be confused with e.g. "Correlation of Fixed Effects"in lmer output).